

SQL Notes

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Table: Patients

Column Name	Type
patient_id	int
patient_name	varchar
conditions	varchar

Write a solution to find the patient_id, patient_name, and conditions of the patients who have Type I Diabetes. Type I Diabetes always starts with DIAB1 prefix. Return the result table in any order.

```
# Write your MySQL query statement below
SELECT *
FROM patients
WHERE conditions REGEXP '(^| )DIAB1';
```

Table: Employees

Column Name	Type
employee_id	int
name	varchar
salary	int

Write a solution to calculate the bonus of each employee. The bonus of an employee is 100% of their salary if the ID of the employee is an odd number and the employee's name does not start with the character 'M'. The bonus of an employee is 0 otherwise.

```
SELECT employee_id, CASE
    WHEN employee_id%2=0 THEN 0
    WHEN name LIKE 'M%' THEN 0
    ELSE salary
    END
    AS bonus

FROM Employees
ORDER BY employee_id;
```

Table: Employee

Column Name	Type
id	int
salary	int

Write a solution to find the second highest distinct salary from the Employee table. If there is no second highest salary, return null

```
SELECT(  
    SELECT DISTINCT salary  
    FROM Employee  
    ORDER BY salary DESC  
    LIMIT 1 OFFSET 1  
) AS SecondHighestSalary;
```

OR

```
SELECT MAX(salary) AS SecondHighestSalary  
FROM Employee  
WHERE salary <(  
    SELECT MAX(salary)  
    FROM Employee  
); (not as efficient)
```

Tables

Customers

Column	Type
id	int
name	varchar

Orders

Column	Type
id	int
customerId	int

Find the names of customers who never placed an order.
Output column: Customers

```
SELECT name AS Customers
FROM Customers
WHERE id NOT IN (
    SELECT customerId
    FROM Orders
);
```

OR

```
# Write your MySQL query statement below
SELECT name as Customers
FROM Customers c
WHERE NOT EXISTS (

    SELECT 1
    FROM Orders o
    WHERE c.id=o.customerId
);
```

Table: Users

Column	Type
user_id	int
name	varchar

Each name should be formatted so that:

- First character is uppercase
- Remaining characters are lowercase

Return the result ordered by user_id.

```
# Write your MySQL query statement below
SELECT user_id,

CONCAT(UPPER(SUBSTRING(name,1,1)), LOWER(SUBSTRING(name,2)))
AS name
FROM Users
ORDER BY user_id;
```

Tables

SalesPerson

Column	Type
sales_id	int
name	varchar
salary	int
commission_rate	int
hire_date	date

Company

Column	Type
com_id	int
name	varchar
city	varchar

Orders

Column	Type
order_id	int
order_date	date
com_id	int
sales_id	int
amount	int

Find the names of all salespersons who did not have any orders related to the company "RED".

```
SELECT name
FROM SalesPerson
WHERE sales_id NOT IN

(
    SELECT sales_id
    FROM Orders
    WHERE com_id=(
        SELECT com_id
        FROM Company
        WHERE name='RED'
    )
)
```

```
)  
);
```

OR

```
SELECT name  
FROM SalesPerson s  
WHERE NOT EXISTS(  
    SELECT 1  
    FROM Orders o  
    JOIN Company c  
    ON o.com_id=c.com_id  
  
    WHERE s.sales_id=o.sales_id  
    AND c.name='RED'  
);
```